

Internet Of Things

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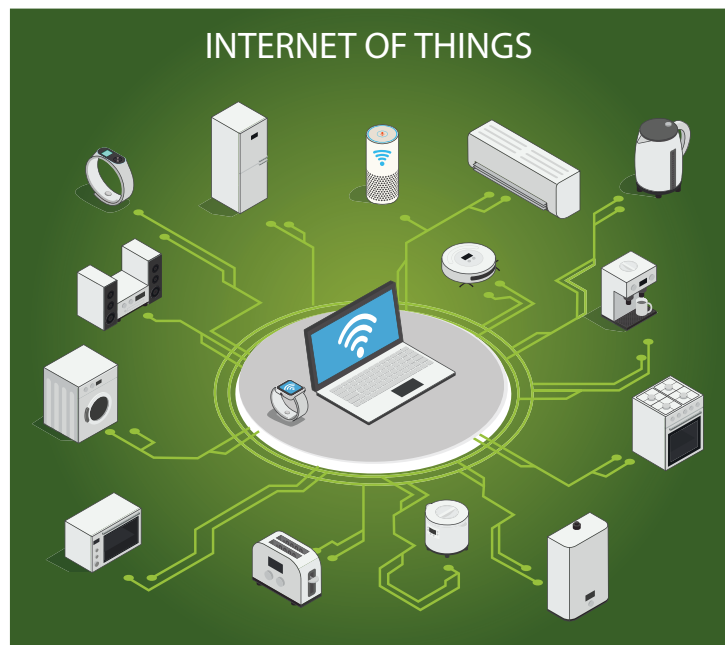
Check out this device I created.
It has an in-built wireless module.

Nice! That means I can control it
remotely with a smartphone app.



When we walk into a room, we use switches to turn on lights, fans, TVs, and other appliances. They can be turned on automatically through an application. The app is controlled via the internet. This is called automation. It is an upcoming trend which works on the “Internet of Things”.

- The Internet of things (IoT) is a network of physical objects or “things” embedded with sensors, software, and other technologies. IoT connects devices and systems to exchange data over a network.
- Internet of Things implies that the devices we use in our daily lives will have the ability to communicate with each other.
- The ‘Thing’ in IoT refers to any device which has a connected sensor to collect and transfer data over a network without human interference.



- IoT connects simple devices like refrigerators to a computer network. It uses software to connect them to us through a smartphone.
- Network connections include Bluetooth, Wi-Fi, and Near-Field Communication (NFC).
- Connected devices work as and when we want to use them. They receive and share information quickly, which improves efficiency.

Examples of IoT

Simple electronic appliances like refrigerators, ACs, cars and washing machines can include internet access through wireless networks.

For example: A simple refrigerator can become 'smart' by using an app to scan groceries and display their expiration date. This app would be able to send emails and notifications before a product expires.

Components of an IoT System



We can build a basic IoT system with the help of the following:

- 1. Devices and Sensors:** Sensors and devices collect data from the surrounding environment.
For example: A temperature reading of our surroundings.
- 2. Connectivity:** Sensors and devices can connect to the internet or a computer through networks like cellular, satellite, Wi-Fi, Bluetooth, etc.
- 3. Data Processing:** Once data from the sensor/device is received by a computer, the computer uses a software/program to process it.
- 4. User Interface:** A user interface is used to display the result of the processed data to the user.
With a user interface, we can interact with a device to turn any application on or off.

Depending on the IoT application, users can control the application.

For example, a user can remotely change a refrigerator's temperature through an app on their phone.

However, in some IoT applications, actions can be performed automatically instead of waiting for an input from the user. For example: If a frozen product is melting in the freezer, the fridge will adjust its temperature based on predefined rules.

Bytes

According to predictions the number of internet-connected IoT devices all over the world will reach or exceed 75 billion by 2025.

Activity

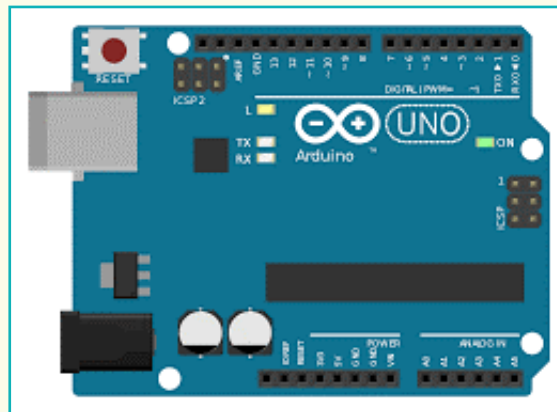
Create a basic IoT application to control an LED through an android app.
We will divide the process into 4 steps:

- 1 Build an electronic application with necessary electronic components and Arduino as a processor.

Recall:

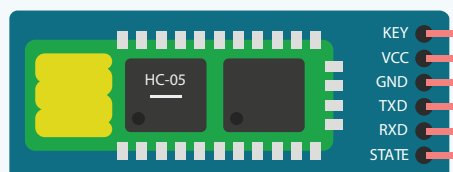
a. Arduino Board:

- Arduino is a tool to develop interactive devices that take inputs from different sensors or devices. They process the data and send the output to applications or output devices.
- Arduino is an open-source platform based on an easy-to-use hardware and software.
- The hardware is a processing board, also known as an Arduino board. It can be programmed using software to perform tasks.
- There are many Arduino boards available. The most popular board is the Arduino UNO board.



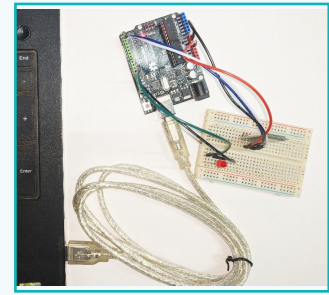
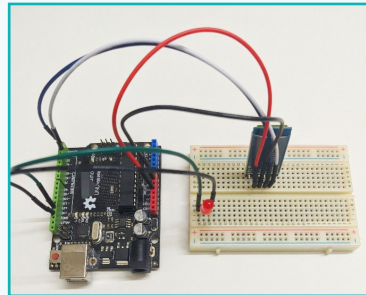
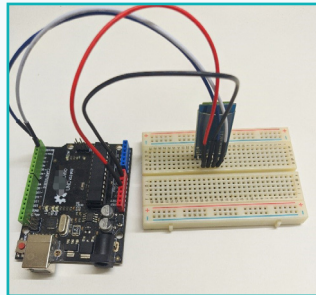
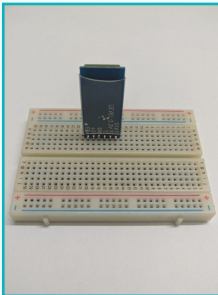
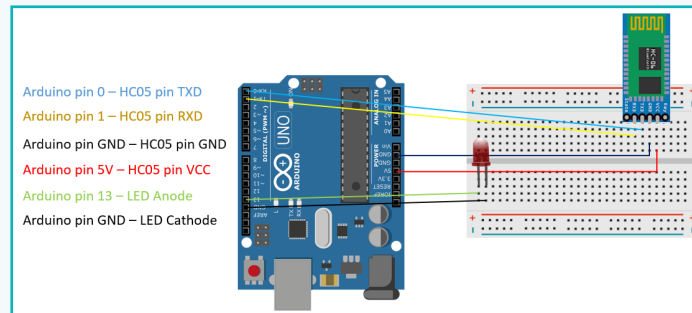
b. Bluetooth Module:

A Bluetooth module is a wireless communication device used to communicate between two processors. These could be Arduinos or any device with Bluetooth functionality, like a smartphone or laptop.



Activity

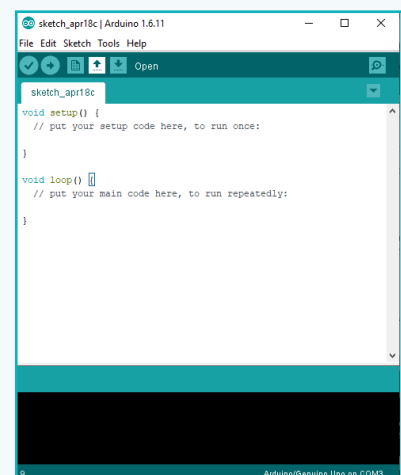
c. Electronic Application Connection



- 2 Create a program in Arduino IDE that processes input received from the user and controls the LED

a. Arduino IDE:

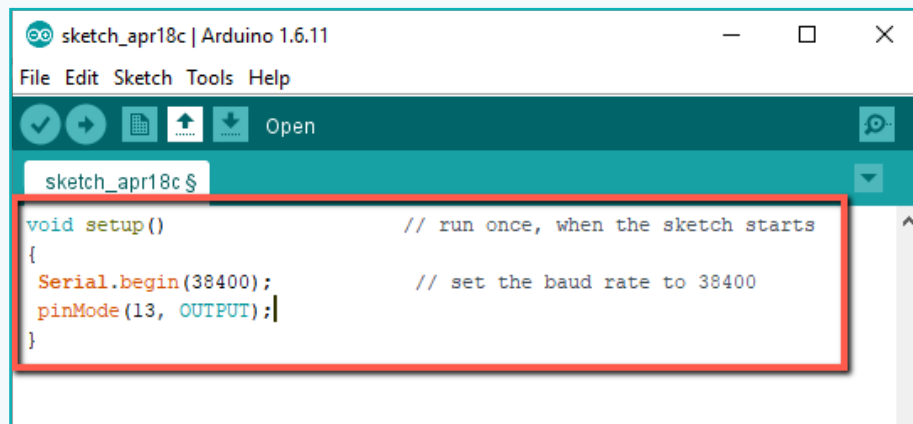
- Arduino IDE (Integrated Development Environment) is the most common software used to program Arduino boards.
- The Arduino programming language is C++, with the addition of methods and functions.
- A program written in Arduino programming language is called sketch. It is saved with the extension .ino.



Activity

b. Arduino IDE program

- Set the digital pin 13 in output mode, and the baud rate to 38400.



```
sketch_apr18c | Arduino 1.6.11
File Edit Sketch Tools Help
sketch_apr18c $
void setup() // run once, when the sketch starts
{
  Serial.begin(38400); // set the baud rate to 38400
  pinMode(13, OUTPUT);
}
```

- Check if data is available on the Bluetooth pin.

```
void loop()
{
  if(Serial.available()){
    while(Serial.available())
    {
      char inChar = (char)Serial.read(); //read the input
      inputString += inChar; //make a string of the characters coming on serial
    }

    Serial.println(inputString);
    while (Serial.available() > 0)
    { junk = Serial.read() ; } // clear the serial buffer
  }
}
```

- Check if the data received is 0 or 1. If it is 0, turn the LED off and if it is 1, turn the LED on.

```
Serial.println(inputString);
while (Serial.available() > 0)
{ junk = Serial.read() ; } // clear the serial buffer

if(inputString == "1"){ //in case of '1' turn the LED on
  digitalWrite(13, HIGH);
}else if(inputString == "0"){ //incase of '0' turn the LED off
  digitalWrite(13, LOW);
}
inputString = "";
}}
```

Activity

c. Final code

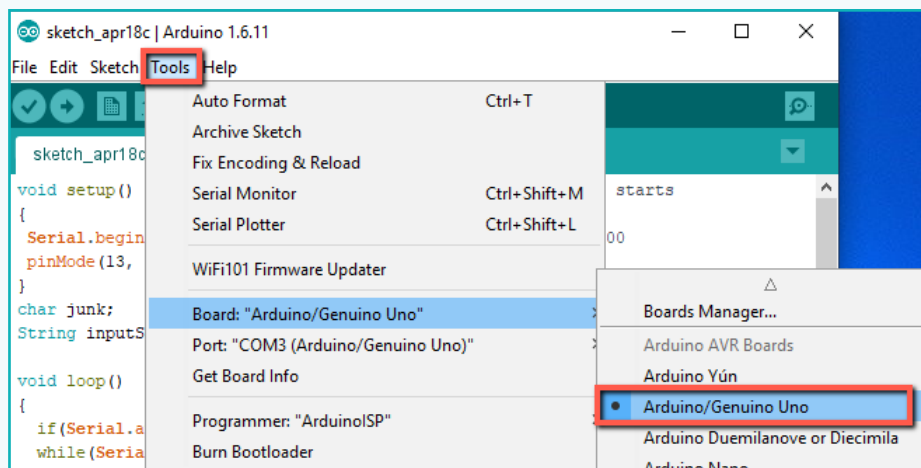
```
void setup()                // run once, when the sketch starts
{
  Serial.begin(38400);      // set the baud rate to 38400
  pinMode(13, OUTPUT);
}
char junk;
String inputString="";

void loop()
{
  if(Serial.available()){
    while(Serial.available())
    {
      char inChar = (char)Serial.read(); //read the input
      inputString += inChar; //make a string of the characters coming on serial
    }
    Serial.println(inputString);
    while (Serial.available() > 0)
    { junk = Serial.read() ; } // clear the serial buffer

    if(inputString == "1"){ //in case of '1' turn the LED on
      digitalWrite(13, HIGH);
    }else if(inputString == "0"){ //incase of '0' turn the LED off
      digitalWrite(13, LOW);
    }
    inputString = "";
  }
}
```

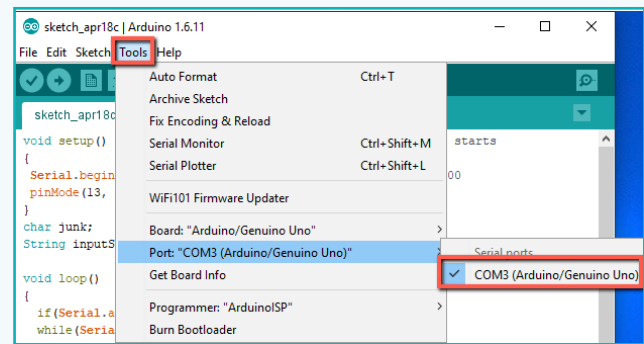
d. Upload program on to Arduino

- Use a USB cable to connect the Arduino Uno board to the computer.
- Select the Arduino Uno board from the Tools menu.

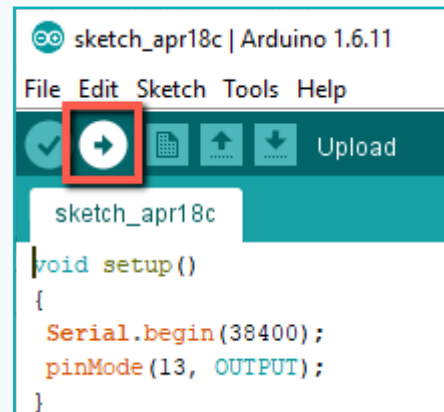


Activity

- From the Tools menu, select the communication port where the Arduino board is connected to the computer.

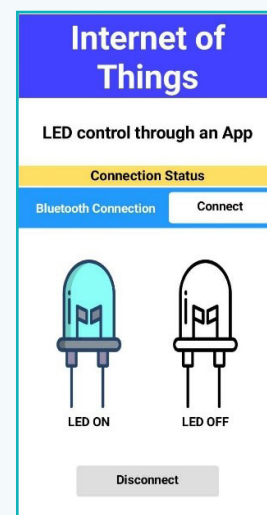


- Click the upload button to upload the code.



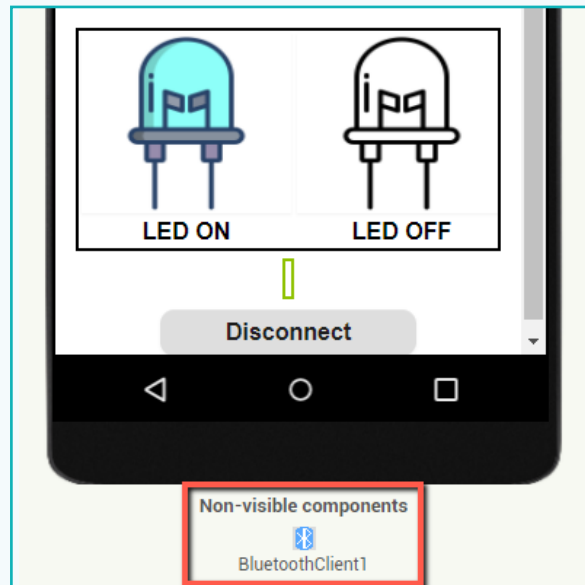
- Design an app user interface (UI) in MIT App inventor 2. The user will interact with the electronic application through this.

a. UI design

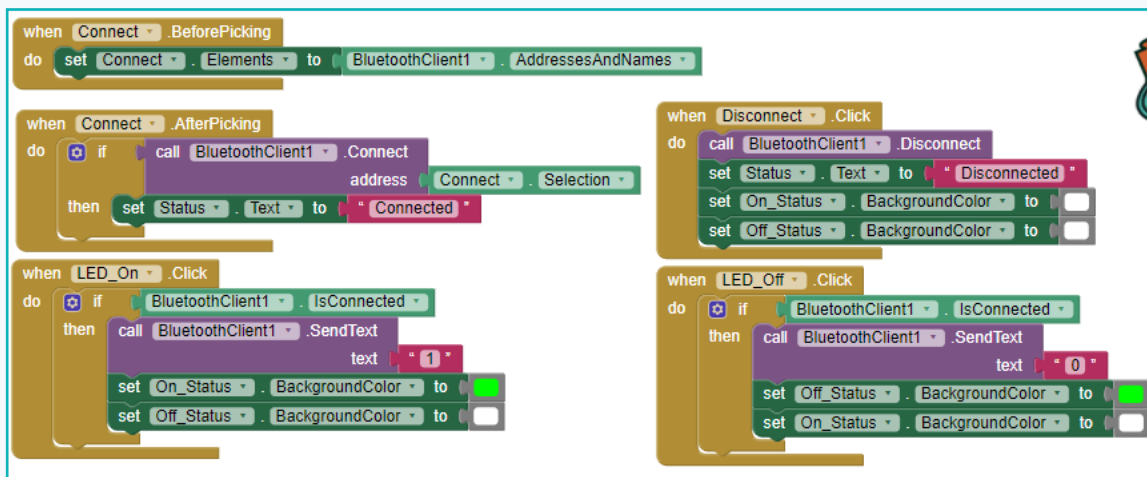


Activity

b. Add Bluetooth client component in UI.



4 Implement user interface functionality in MIT app inventor 2.



Exercise

State whether True or False

- 1 The 'Thing' in IoT can be any device with a connected sensor that collects and transfers data over a network without human interference.

▲ _____

- 2 A computer uses a software/program to process data once it is received from the sensors.

▲ _____

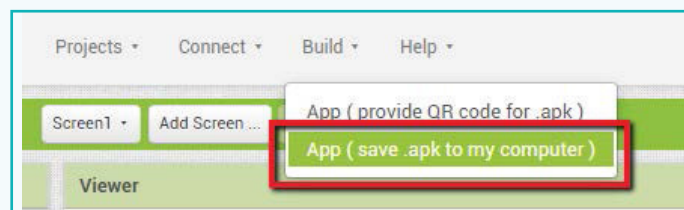
- 3 Bluetooth module is a wired communication device used to communicate between two processors.

▲ _____

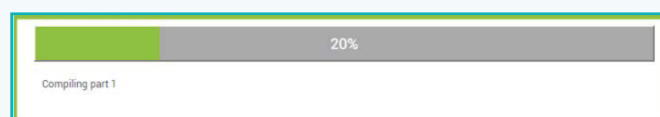
Activity

Test

- Open MIT App Inventor and click on “Build” at the top of the designer screen. Then click on “App (save .apk to my computer)” inside Build.

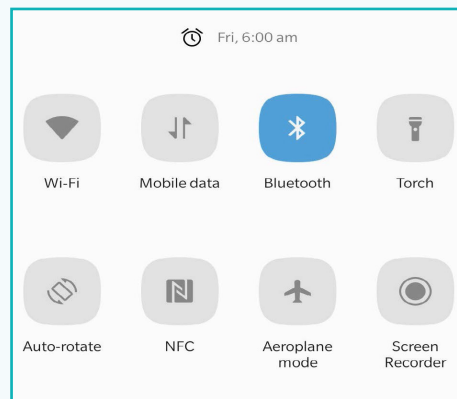


- Wait for the process to be 100% complete.

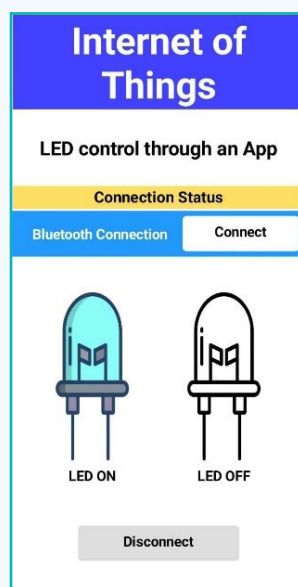


Activity

- Copy the downloaded application from the “downloads” folder and paste it in 2 Android phones.
- When we open the file on a phone, we see an option to install the app. Click on “Install” and repeat the process on the other phone.
- Open the applications on both smartphones.
- Enable Bluetooth on the devices.

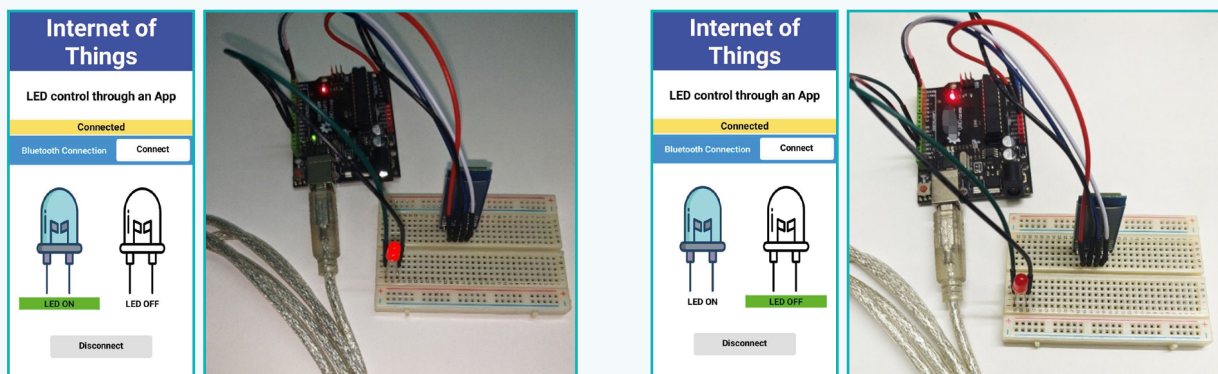


- Press “Connect” and select the Bluetooth module network.

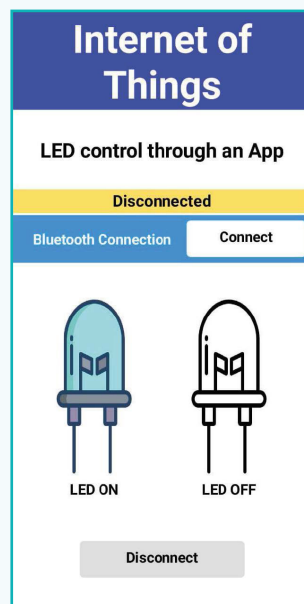


Activity

- Once the app is connected with the electronic application via Bluetooth, we can test it. We need to check if the LED turns ON and OFF when the respective buttons are pressed on the app.



- Press the “Disconnect” button to disconnect the electronic app from the device.



Sensors communicate with a computer via a network connection. Once data reaches the computer, the software processes it and then performs an action. This could be sending an alert to the user or automatically adjusting the sensors/devices according to predetermined instructions.

Different IoT devices can be used for various applications. These devices make our lives easier.



Tech Flash

- **Internet of Things** : The Internet of Things (IoT) is the network of physical objects or “things” embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems.
- **Bluetooth** : Bluetooth is an open wireless technology used to transfer data over short distances from a fixed or mobile device.
- **Arduino Board** : Arduino is a tool to develop interactive devices that takes inputs from different sensors or devices, processes the data, and gives output to applications or output devices.
- **Arduino IDE** : Arduino IDE (Integrated Development Environment) is the most common text-based software used to program Arduino boards.